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Compound Inequalities

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Compound Inequalities

A **compound inequality** combines two statements. You can use inverse operations to solve compound inequalities. Whatever you do to one part of the inequality, you must do to every part.

A graph of the solution to a compound inequality makes it easier to understand the answer. To make a graph, draw a short line and mark the important numbers from the solution. Then shade the part of the line that contains the solution set.

Example 1

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$$-6 < x + 3 < -4$$

Subtract 3 from each part of the inequality.

$$-6 - 3 < x + 3 - 3 < -4 - 3$$

$$-9 < x < -7$$

The solution is $-9 < x < -7$.

The graph of the solution is:



Example 2

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$$a + 4 \geq 10 \text{ or } a - 1 < -5$$

Solve the inequalities separately.

$$a + 4 \geq 10$$

$$a - 1 < -5$$

$$a + 4 - 4 \geq 10 - 4 \qquad a - 1 + 1 < -5 + 1$$

$$a \geq 6$$

$$a < -4$$

The solution is $a \geq 6$ or $a < -4$.

The graph of the solution is:



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Lesson 5: Inequalities

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